



TACAN / HSI Navigation

by NARC



TACAN / HSI Navigation



Basic HSI and TACAN operation, to include tuning and identifying TACANs and basic HSI symbols, will not be covered in this document. Please see the DCS A-10C user manual for reference to those topics.

Overview

- TACAN homing navigation
- Direct to the station navigation
- Intercepting a TACAN radial
- Point to point navigation



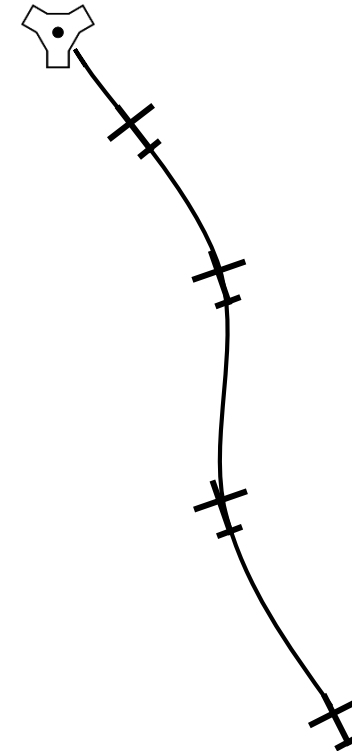
TACAN / HSI Navigation



TACAN Homing

- A basic, **non wind** corrected method to proceed to or away from a TACAN
- Does not follow a direct **ground track** to the station
- Procedure
 - Tune / Identify / Twist
 - When CDI drifts, **twist** in a new course to center CDI
 - Continue until overflight of TACAN
- Not advised for use in the jet as this method does not proceed direct to the station

TACAN Homing Ground Track





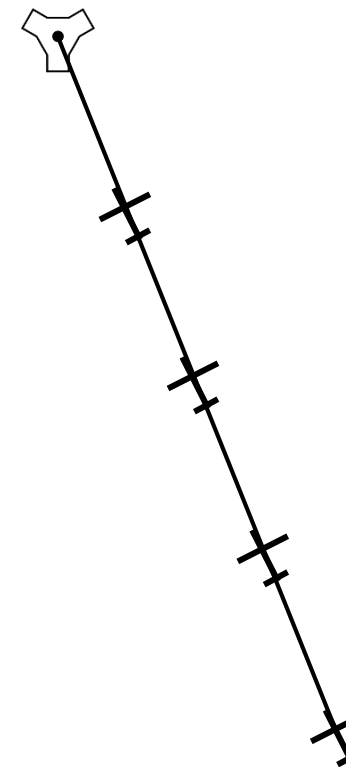
TACAN / HSI Navigation



Direct to the Station

- A basic, **wind** corrected method to proceed direct to or away from a TACAN
- This method will follow a direct **ground track** to the station
- Procedure
 - Tune / Identify / Twist
 - When CDI drifts, change the A/C heading to fly into the deflected CDI
 - Hold a crab angle to keep the CDI centered
- This method should be used when flying direct to a station

TACAN Direct to Ground Track



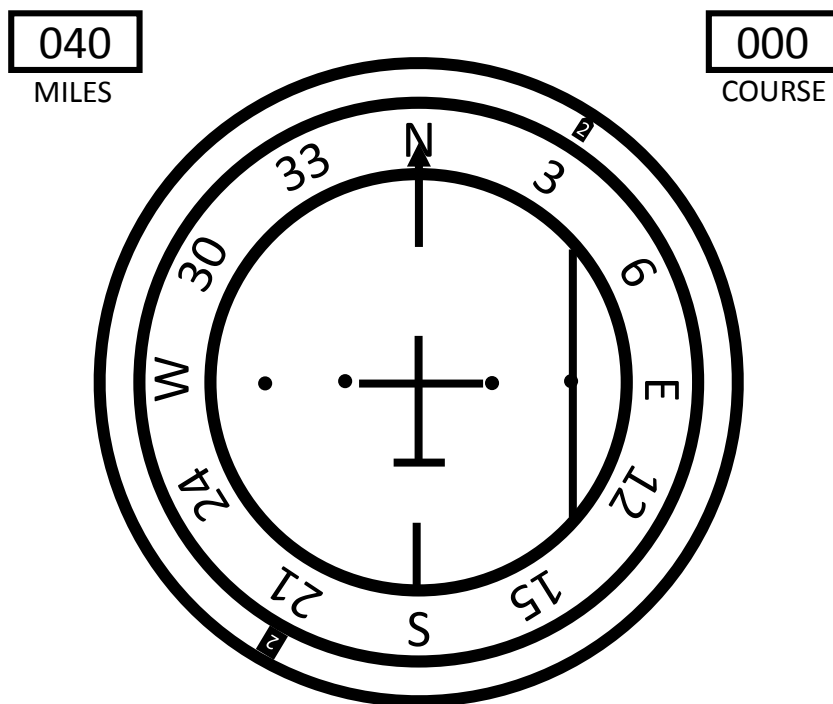


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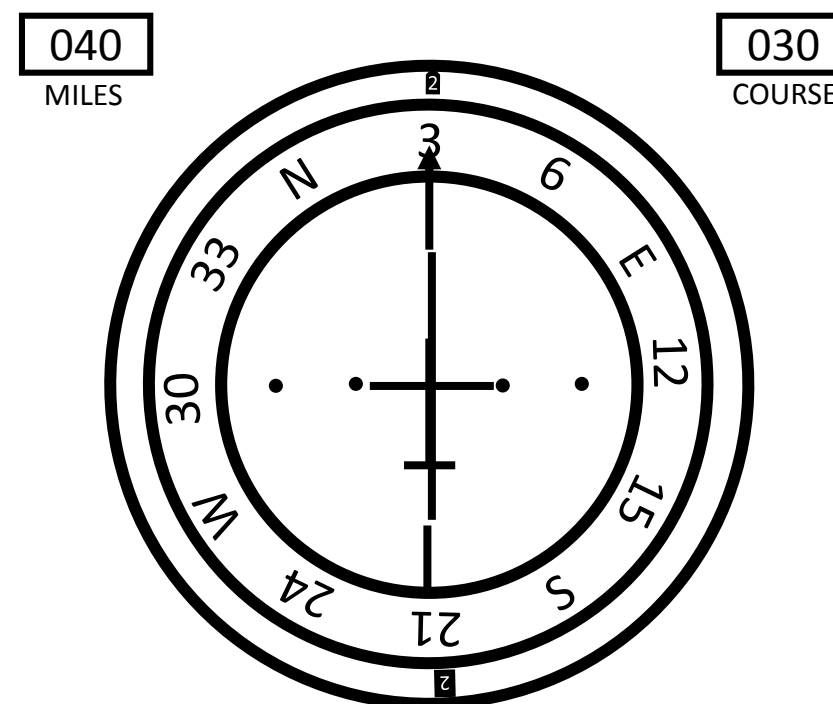


Direct to the Station (continued)

- Suppose you **Tune** up Vaziani's TACAN while heading North and you see the following on your HSI:
 - You are currently on the 210 radial for 40 miles
 - You decide you want to fly direct to the station



- **Twist** in the inbound course with the course set knob (030)
- The CDI will now center up
- Turn to a heading of 030
- If there was no wind you could fly a heading and a course of 030 direct to the station
- See the next page for a wind correction example





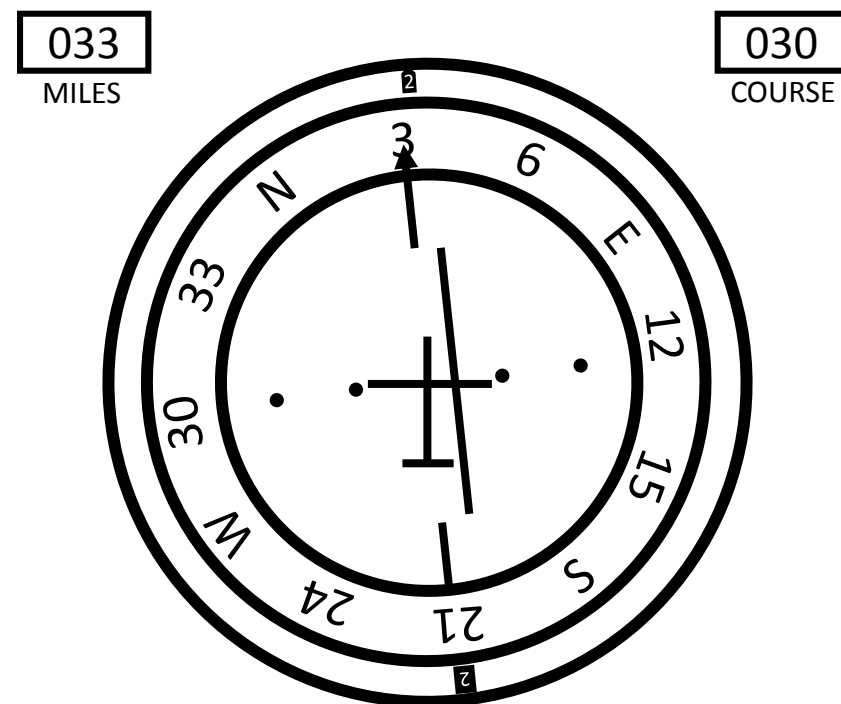
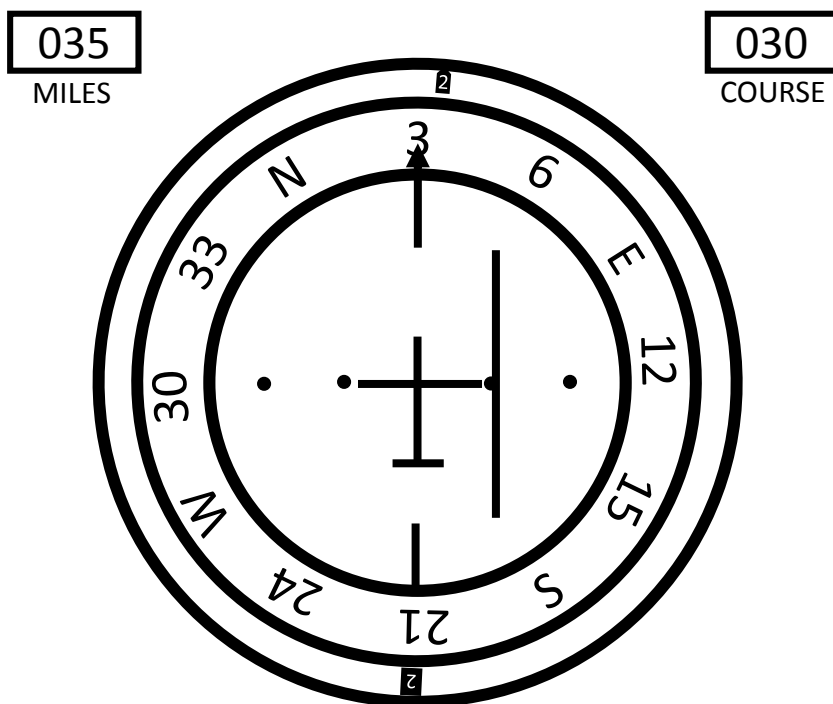
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Direct to the Station (continued)

- After flying a heading of 030 for 5 miles you notice the CDI has drifted to the right approximate 5 degs, as shown below
- It is apparent there is a right to left wind blowing off the 030 course line direct to the station
- You want to continue to fly direct and not home our way to the TACAN

- You also don't want to keep correcting for the wind the whole way back
- To find the proper crab angle, turn into the CDI **3 times** the amount of deflection
- In this case turn right to a heading of 045
- Once the CDI starts correcting back left decrease your corrective heading until the CDI stabilizes
- Note your heading and the difference between the course you want (030) and your current heading.
- In this case we are at a heading of 035, so we know our crab is 5 deg right



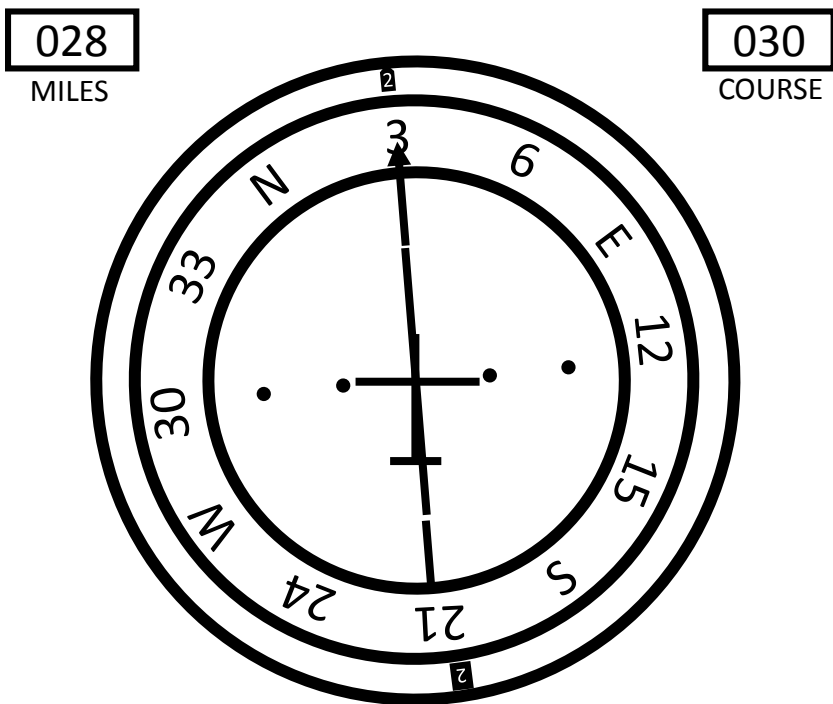


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Direct to the Station (continued)

- Now that we have the crab angle of 5 deg, fly a heading greater than the crab angle to finish centering the CDI on the 030 course inbound
- Once centered reset the crab angle of 5 deg right as shown below

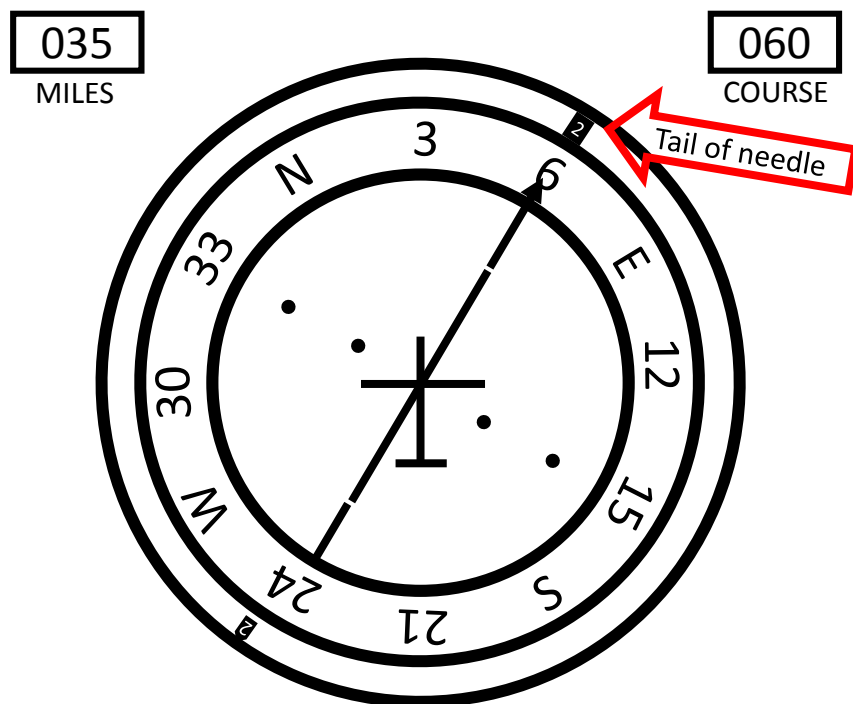




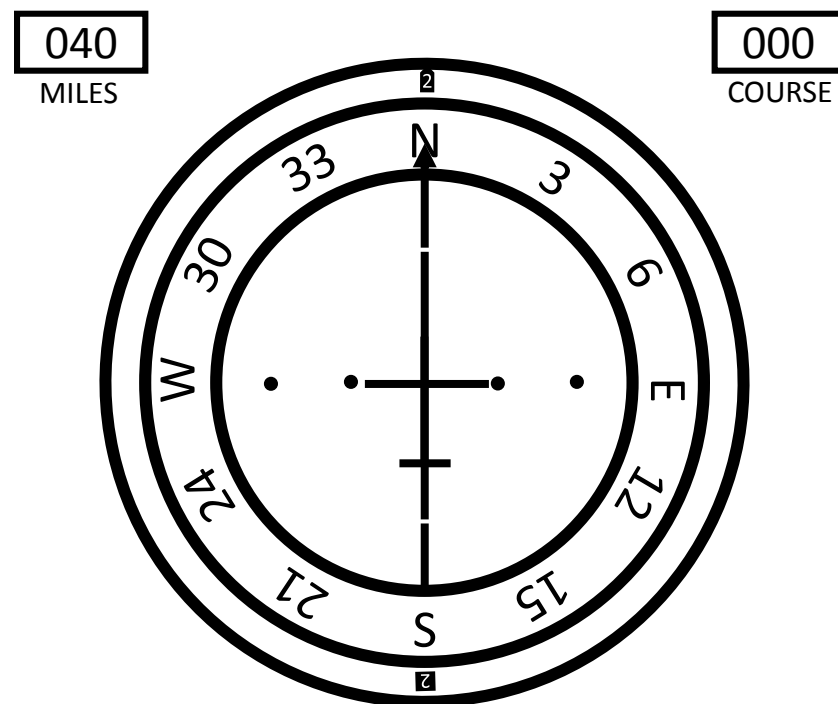
TACAN / HSI Navigation

Intercept a TACAN Radial

- **Radial** – these are the radio signals that emit outward from the TACAN in all directions (360°)
- **No matter if you are going toward or away from the TACAN, the TAIL of the TACAN needle will tell you what radial you are currently on**
- Below you are headed NE away from the station on the 060 radial for 35 nm



- Say we want to intercept the Vaziani 252 Radial and then fly inbound to BAGEM
- We will start to the S on the 180 radial at 40 nm
- You would have a display similar to the one below





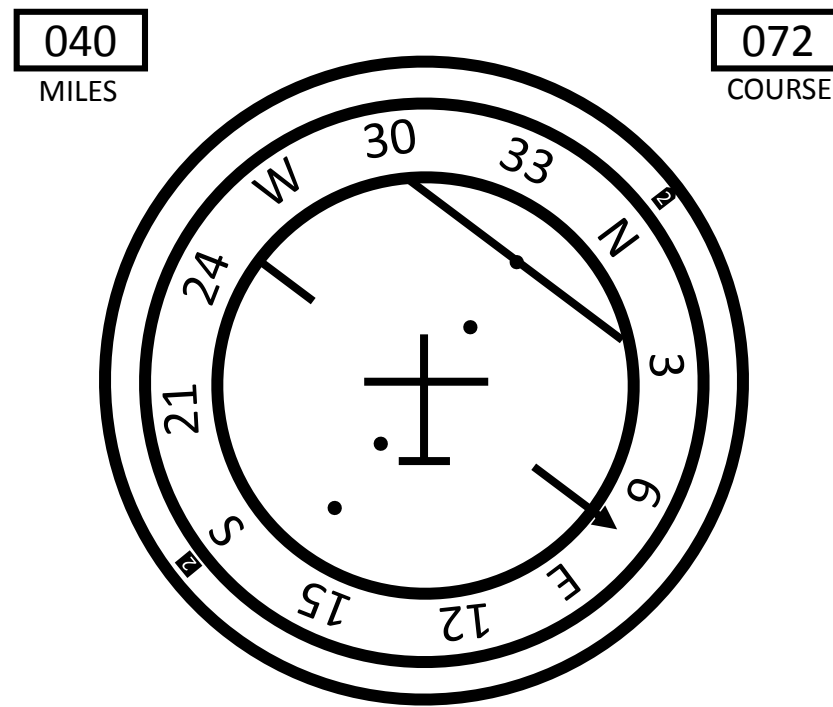
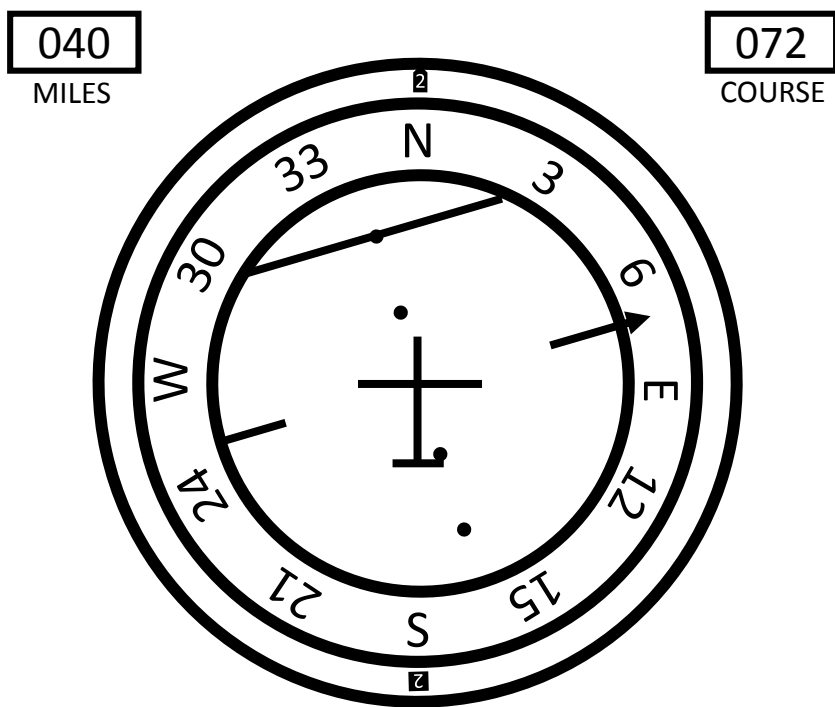
TACAN / HSI Navigation



Intercept a TACAN Radial (continued)

- Since we want to fly on the 252 radial inbound to intercept BAGEM, we will turn our COURSE knob to the **INBOUND** course or reciprocal of the radial
- From the display below, we can see that we need to head NW to push the needle to the right
- The tail of the needle will then begin to track left towards the 252 radial

- Since we were on the 180 radial, we need to push the tail left 72 radials (252-180)
- The heading offset to use is a factor of how close you are to the TACAN and how many radials you need to cross
 - Far from TACAN and large number of radials – Use 40-60 deg
 - Mid range and medium number of radials – Use 30-40 deg
 - Close range and small number of radials – Use 15-30 deg
- Experience will help decide how aggressive you want to be in a given situation
- For this example we will use 50 degs so a heading of 310





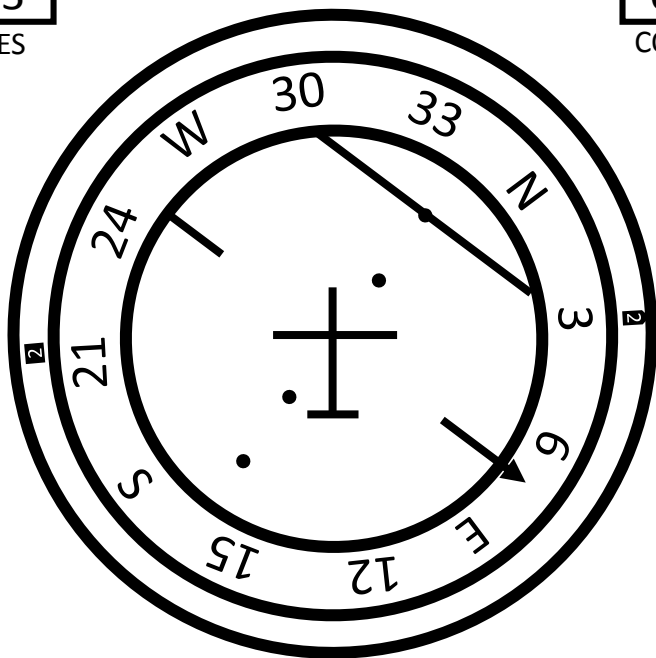
TACAN / HSI Navigation

Intercept a TACAN Radial (continued)

- As we continue on the 310 heading, the TACAN needle Head will begin falling toward 072 and the TACAN needle TAIL will begin rising toward 252
- Below are depictions of a few changes in the HSI as we fly a heading of 310
- Initial the DME will decrease until we get abeam and then it will increase

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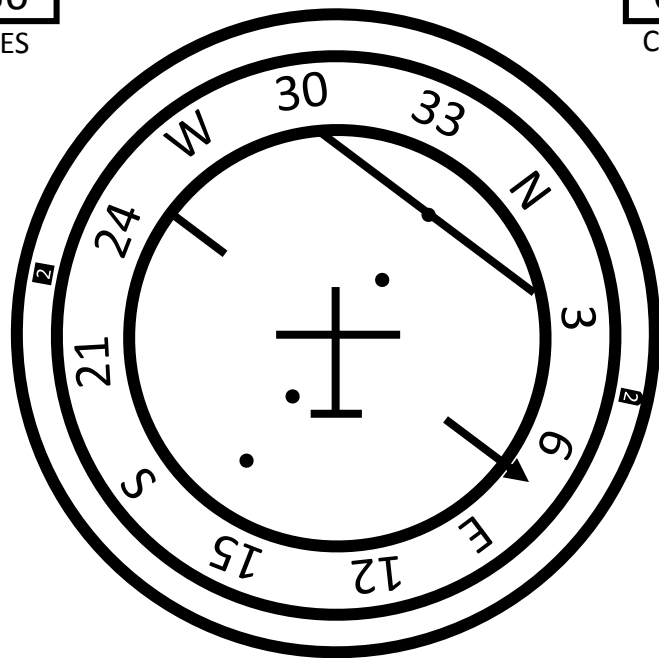


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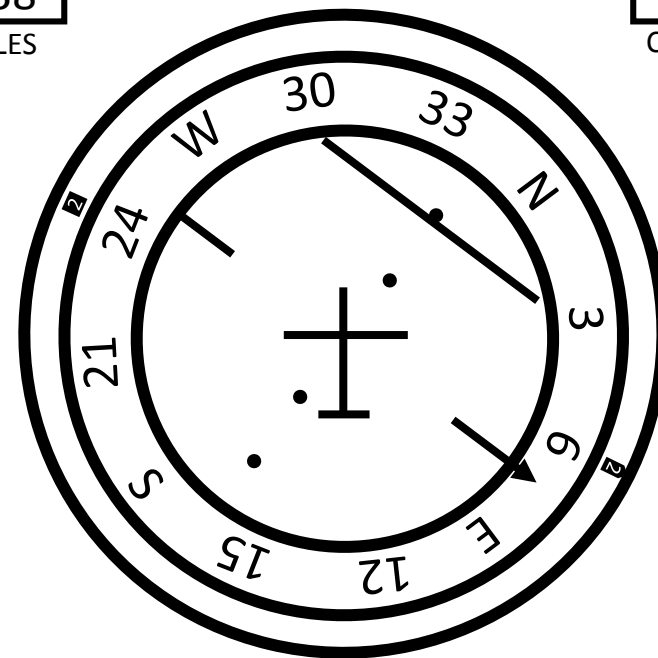


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TACAN / HSI Navigation



Intercept a TACAN Radial (continued)

- As we get closer to intercepting the radial we want to intercept we need to think about how to lead our turn
- A lead point in nautical miles can be calculated using the 10% rule and the 60 to 1 rule
- Both rules assume no greater than a 90 deg turn to intercept
- 10 % Rule – 10% of the A/C **Ground Speed** = the amount of lead in nautical miles
 - For Example: My IAS shows 250 for the approach, but my GS shows 260.
 - Lead point for intercept will be 2.6 nm
- 60 – 1 Rule

DME	NM between radials
60	1 nm
50	5/6 nm
40	2/3 nm
30	1/2 nm
20	1/3 nm
10	1/6 nm

- We're at approximately 40 nm so there are 2/3 nm per radial.
- Our lead point will be approximately 4 radials



TACAN / HSI Navigation

Intercept a TACAN Radial (continued)

- Now that we have our lead radial calculated we want to set that 90 deg intercept geometry as shown by the figure on the left below
- The outer most dot on the HSI represent 10 degs of CDI deflection and the inner 5 degs
- When the CDI gets just inside the 5 deg dot we should start our inbound turn
- The turn assumes a standard rate turn (level 30 deg angle of bank turn)

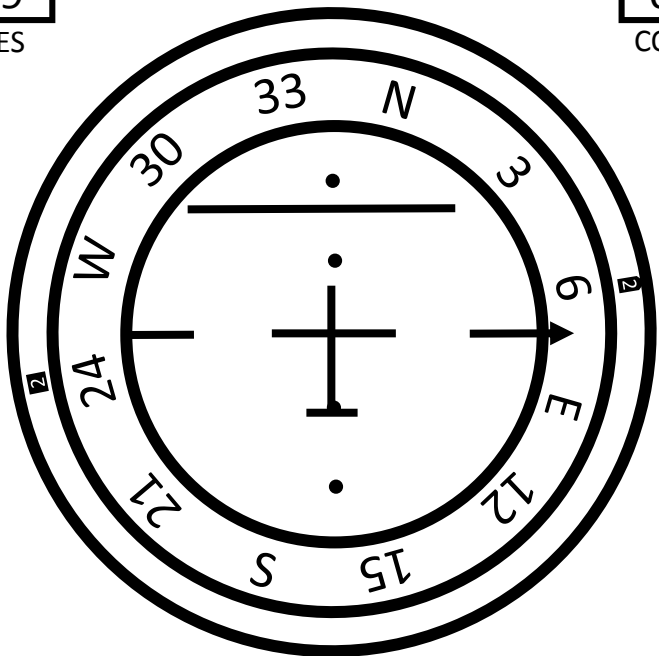
- 90 deg to intercept
- HDG 342

- Begin Turn

- Rolled out on 072 course inbound (252 radial)

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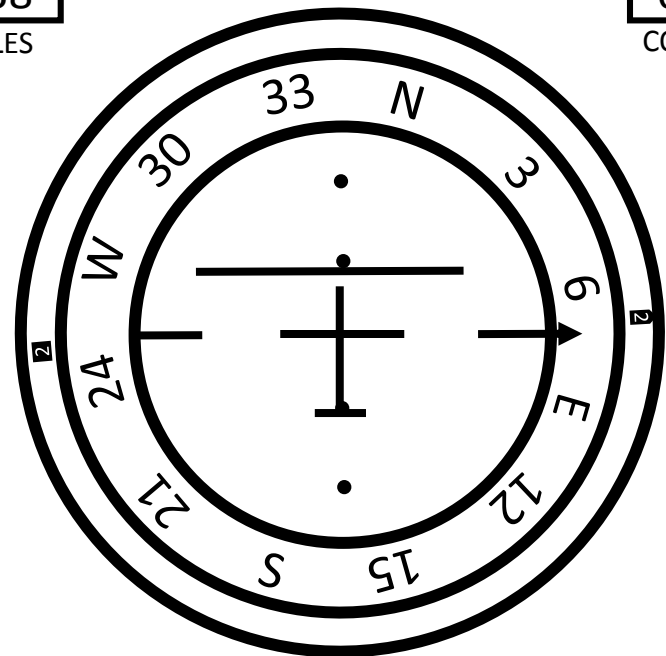


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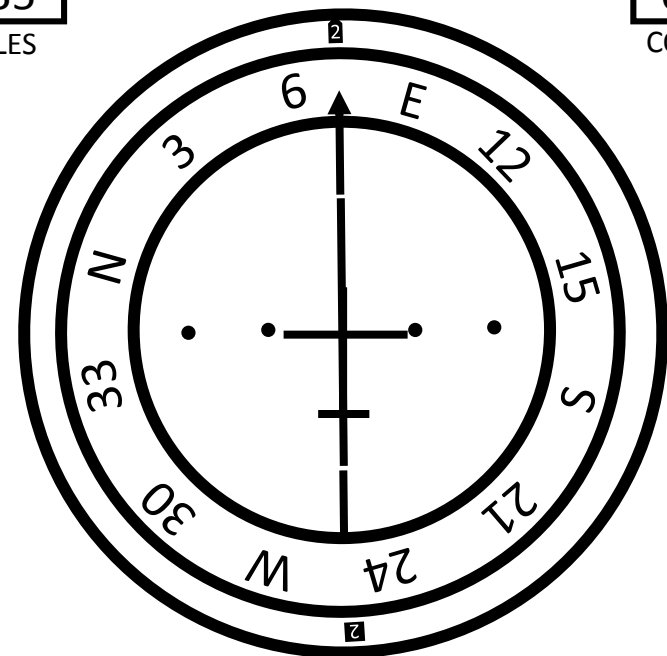


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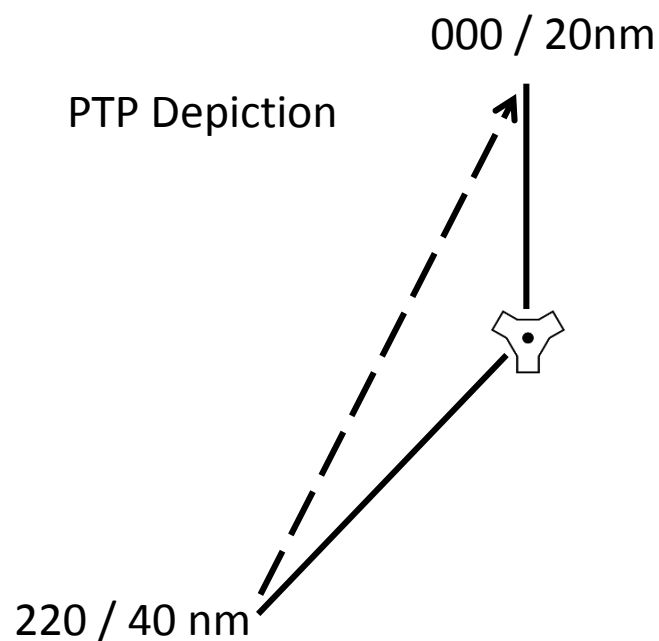


TACAN / HSI Navigation



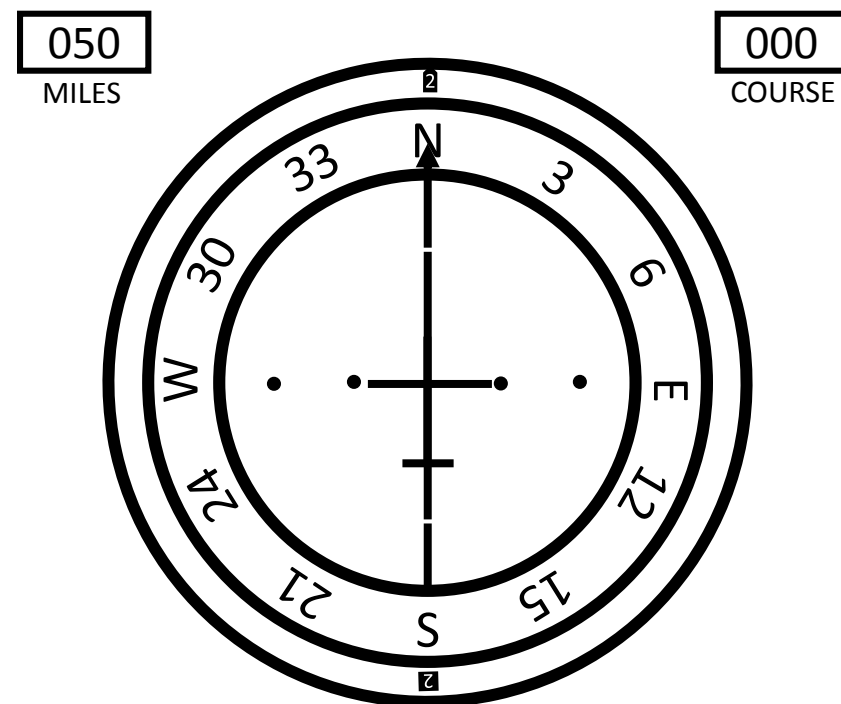
Point to Point TACAN Navigation

- Point to point navigation is flying from one radial/dme direct to another only using the HSI and TACAN
- PTP Navigation can be very useful when you don't have a WPT for a radial/dme cut off a TACAN that you want to fly to.



PTP Navigation Example

- We will use the same example as before starting on the Vaziani 180 radial at 50 nm
- This time we want to go DIRECT to BAGEM (Vaziani 252 / 25 nm)
- We will use what is called the pencil or finger method to determine the heading needed to fly direct to BAGEM
- See the next few pages



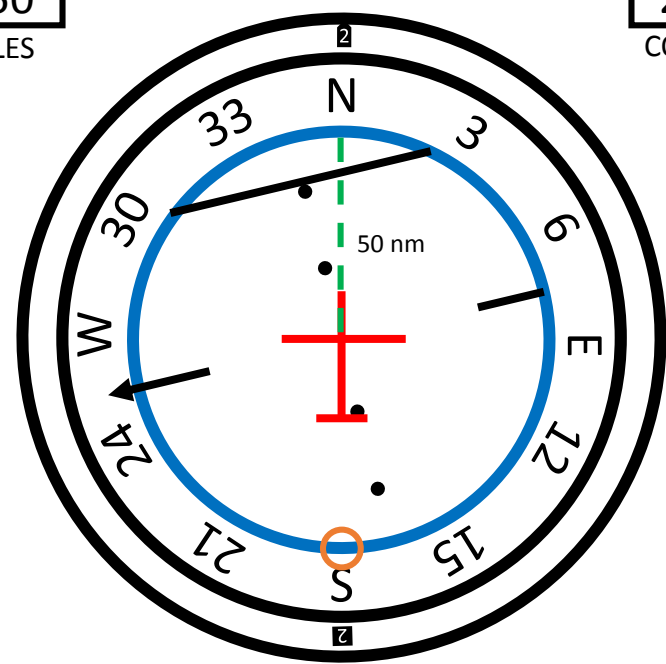


TACAN / HSI Navigation

Point to Point TACAN Navigation (continued)

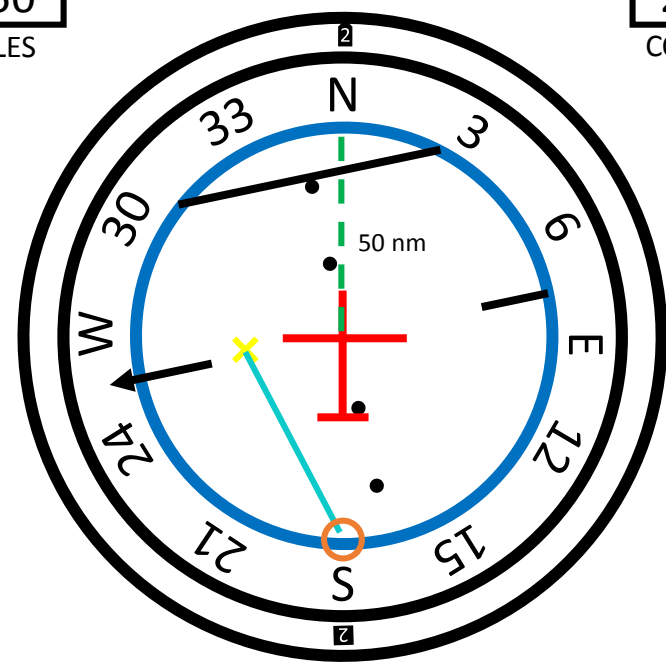
1. The best way to think of this method is that you are plotting where you want to go on the HSI
2. Your current location is the intersection of the TAIL of the needle and the inner circle on the HIS (orange circle)
3. The TACAN location is the center of the HSI (red A/C)
4. Use the larger of the DMEs (50 nm) as the radius (dashed green line) of the inner HSI circle (depicted in blue)
5. The green dashed line is our reference unit of measure
6. We want to go to 25 nm so that's a half unit of measure
7. Spin in the radial you want to go to in the course box
8. Visualize the half unit of measure along the radial (yellow X) to the west of the TACAN since we know BAGEM is West on the 252
9. Use a pencil to connect a line (cyan colored line) from intersection of the TAIL of the TACAN and the inner circle (orange circle)
10. Superimpose that imaginary line over the center of the HIS and it will point to your heading
11. In our case the direct heading to BAGEM is roughly 330

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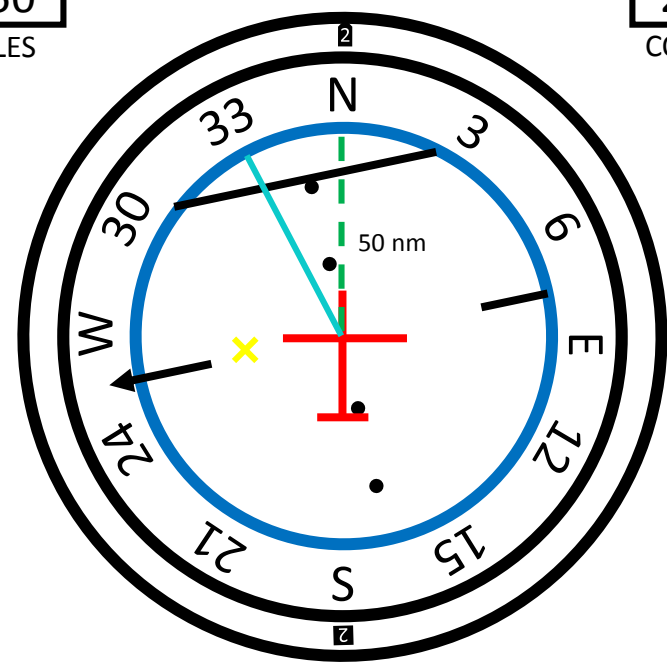
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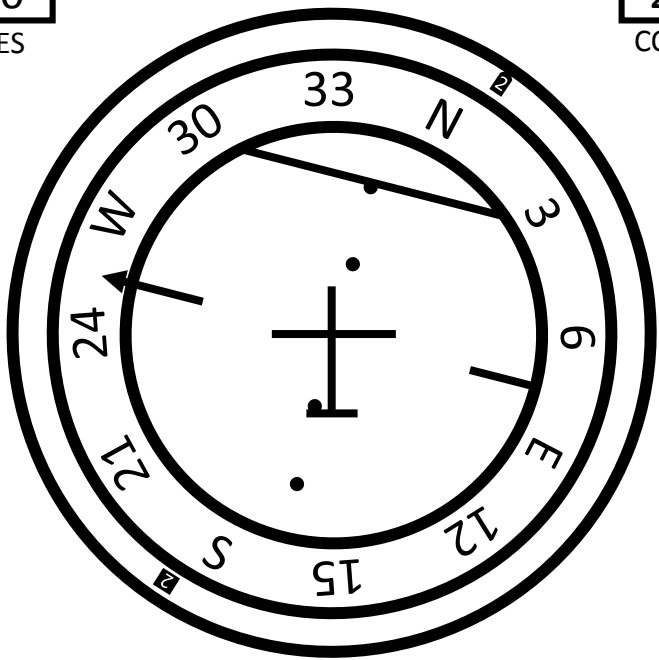
TACAN / HSI Navigation

Point to Point TACAN Navigation (continued)

- As we fly along our initial heading we can update our solution using the pencil method at described before
- Use the HSIs below to practice the pencil method

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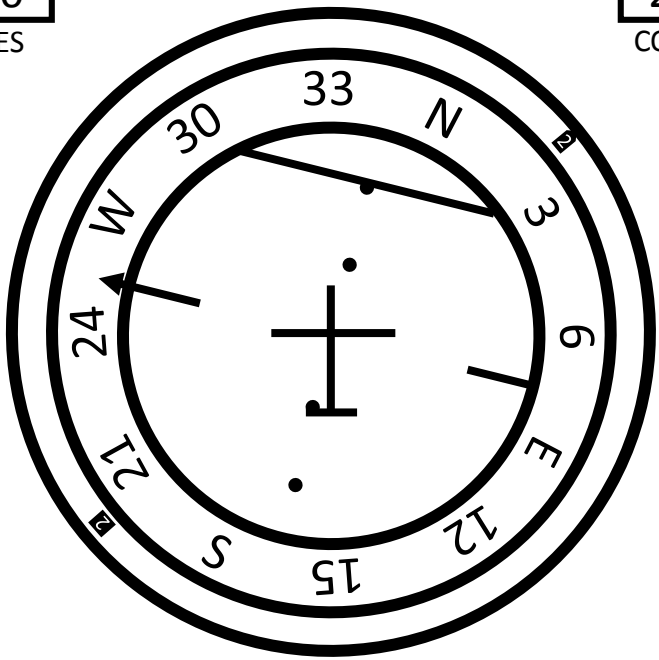


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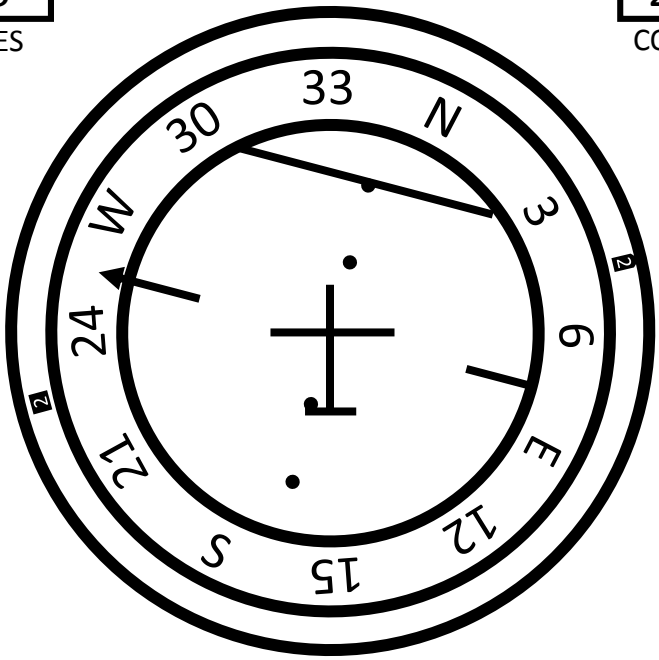


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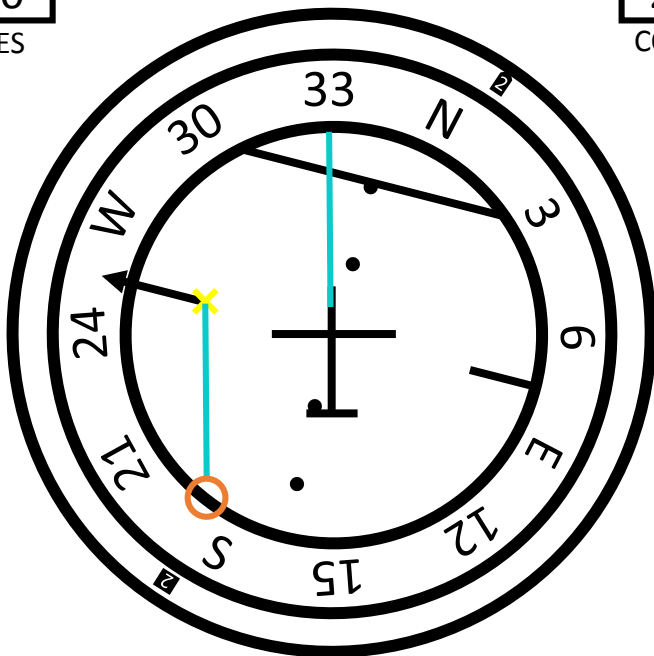
TACAN / HSI Navigation

Point to Point TACAN Navigation (continued)

- Here's the solutions: looks like the initial assessment was pretty good
- The last solution is what is called a CO-DME solution (see the next page for an explanation of CO DME)

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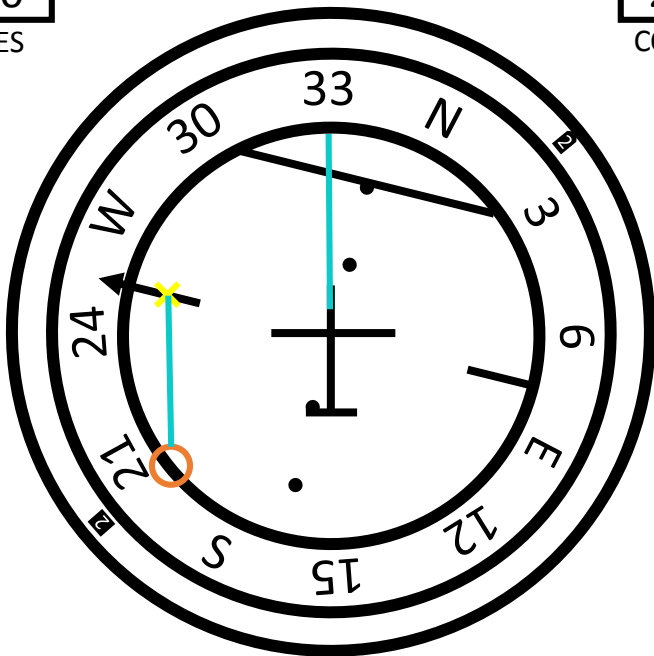


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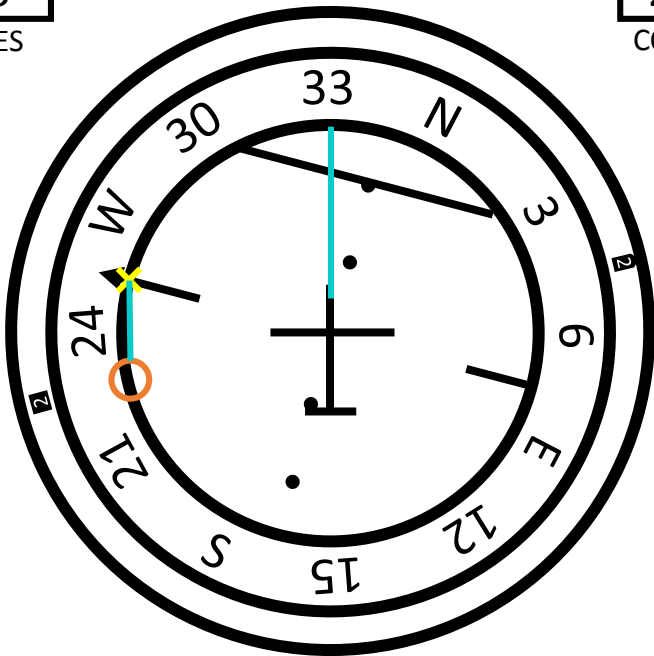


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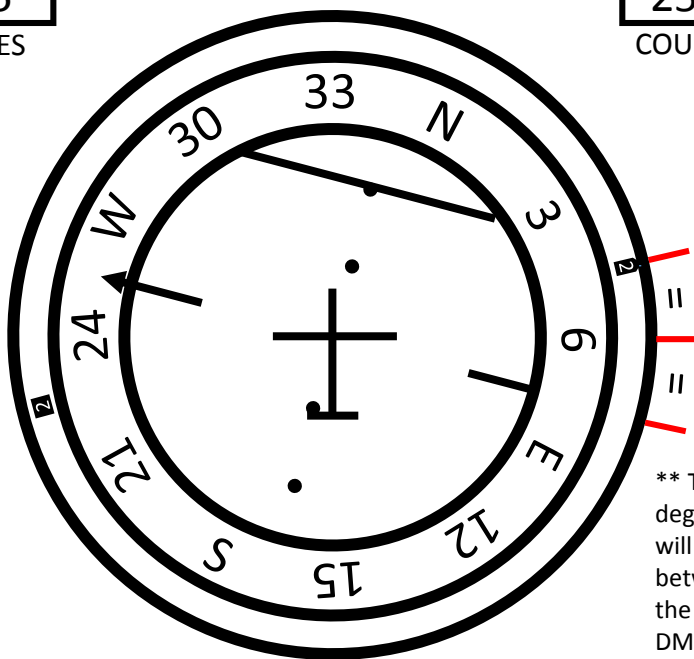
TACAN / HSI Navigation

Point to Point TACAN Navigation (continued)

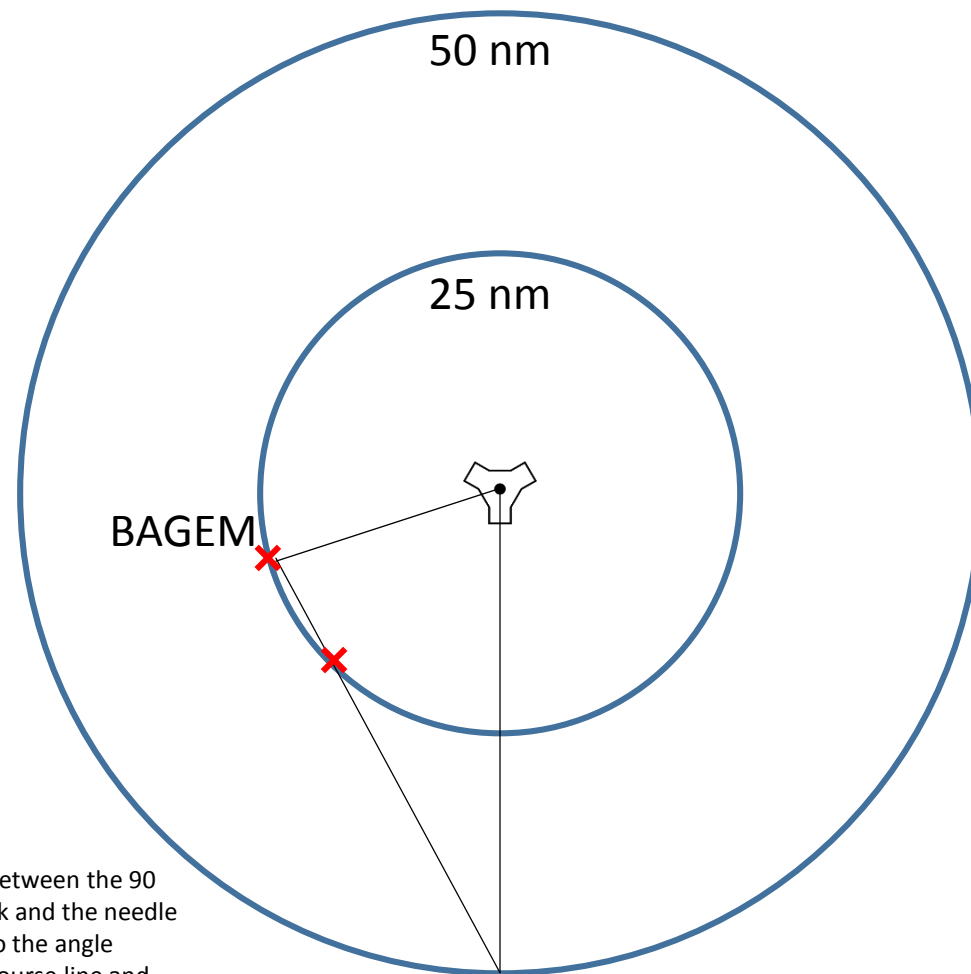
- Here's a top down look at the point to point
- The two red Xs are the CO DME locations for this point to point
- On this particular point to point we intersected the 25 nm arc in two places
 - One at a radial before the one we wanted
 - The second will be at BAGEM

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COURSE



** The angle between the 90 deg benchmark and the needle will be equal to the angle between the course line and the 90 deg benchmark for a CO DME solution





TACAN / HSI Navigation



Point to Point TACAN Navigation (continued)

- Once you have nailed your PTP heading and flown through the point, you should see the CDI centered up with the correct DME in the BOX
- This is depicted below
- If you remember that your location is always at the TAIL of the needle on the inner circle and the TACAN is the center of the HSI, this method will be pretty easy

